STA442 Assignment 2

Due Date: November 6th, 2026, 11:59pm

Instructions:

- **Any question that is labelled with [H] requires a handwritten solution.** This can be written on a piece of paper and scanned, or be written digitally on an iPad or tablet. **If you ignore these instructions, you will be given a grade of 0 for that question.**
- For questions without the label **[H]**, they will require a mix of R-coding and interpretation. RMarkdown is recommended but not required for submission. Include necessary codes and outputs. If your code is complicated, make sure you comment on essential implementations.
- **Please keep your response concise.** Most of the questions can be answered within at most 3-4 sentences. If you are writing an "essay" to answer a question, it is highly likely that some of them are incorrect, which would lead to point deduction.
- Please ensure responses are **legible, and neatly organized**.
- **Submit through Crowdmark**

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1. (a) **[H]** (0.5 pt) Provide a data-generating process where $\hat{\mu}_4$ in Lecture 3 is an unbiased estimator of $\mu = E(y_i)$ (for all i) and $\text{Var}(\hat{\mu}_4) \leq \text{Var}(\bar{y})$. Justify your reasoning.

   (b) **[H]** Does the result in part (a) contradict the Gauss–Markov Theorem? Briefly explain why or why not.

2. **(4 marks)** Consider a study investigating whether high-dimensional health data can be used to predict postpartum depression.

   Suppose we have data from $n = 300$ postpartum patients. For each patient, $p = 2000$ features are derived from wearable sensor measurements, repeated symptom surveys, sleep patterns, activity levels, and other clinical variables.

   Let $y_i = 1$ indicate that patient i develops postpartum depression and $y_i = 0$ otherwise.

   The data are generated as follows:

```
set.seed(442)

y <- ifelse(rnorm(300) > 0.8, 1, 0)

X <- matrix(rnorm(300 * 2000), nrow = 300)
```

   (a) **(1 mark)** What is the best possible classification rate (accuracy) for any machine learning classifier when it is trained by the dataset above and applied to unobserved data from the same population?

   (b) **(1 mark)** Because $p > n$, we may first consider reducing the dimensionality. Compute the correlation between and each gene, and filter the genes with an absolute correlation greater than 0.15.

   (c) **(1 mark)** Apply the 5-fold cross validation with the logistic regression for the dataset reduced in (b), and report the average cross-validation accuracy. You may use cv.glm() function in the boot library in R or build your own cross-validation function.

   (d) **(1 mark)** Argue why the results in (c) is inflated.

3. **[H] (2 marks)** You are involved in a collaborative research project as a statistician with two entomologists. The statistical component of the project involves fitting regression models to large datasets. Together, you have written and submitted a manuscript to an entomology journal.

   (a) **(1 mark)** The manuscript contains a number of scatterplots, each showing an estimated regression line (based on a valid model), associated individual 95% confidence intervals for the regression function at each x value, and the observed data. A referee asks:

   > "I don't understand how 95% of the observations fall outside the 95% confidence interval, as depicted in the figures."

   Briefly explain in two sentences how it is entirely possible that 95% of the observations fall outside the 95% confidence intervals.

   (b) **(1 mark)** Elsewhere in the manuscript, one of the fitted regression models gives a very small p-value ($p < 0.001$) for the slope coefficient.

   A referee comments:

   > "Since the slope is highly statistically significant, this must be a strong and scientifically important relationship."

   Briefly explain in two sentences why this conclusion does not necessarily follow from the reported p-value.

4. **(8 marks)** Suppose that $x_i > 0$ and consider the following data-generating process:

$$Y_i \mid X_i = x_i \sim \text{Poisson}(2 + 1.5x_i),$$

   where the Y_i's are conditionally independent. Recall that for a Poisson random variable,

$$E(Y_i \mid X_i = x_i) = \text{Var}(Y_i \mid X_i = x_i).$$

   One simulated dataset can be generated using the following R code:

```
set.seed(442)

n <- 500
x <- runif(n, 0, 8)
y <- rpois(n, lambda = 2 + 1.5 * x)

plot(x, y)
```

   (a) (1 mark) Suppose we fit the linear regression model

$$Y_i = \alpha_0 + \alpha_1 x_i + \epsilon_i.$$

   What are the true values of α_0 and α_1 that describe the relationship between y_i and x_i? Briefly justify your answer.

   (b) (1 mark) Using the simulated dataset, provide R code to assess the following assumptions of the linear regression model:

   (i) normality of the errors;
   (ii) constant variance;
   (iii) linearity.

   Comment briefly on whether each assumption appears to be satisfied.

(c) (1.5 marks) Carry out a simulation study comparing the bias and varianc of the estimated slope, $\hat{\alpha}_1$, obtained using the following methods:

- ordinary least squares
- weighted least squares with $w_i = \frac{1}{2+1.5x_i}$
- weighted least squares with $w_i = 2 + 1.5x_i$

Use 10,000 simulation replications and a sample size of $n = 100$ in each replication. Report the estimated bias and variance for each estimator and identify which estimator is most efficient in reducting the variance of the estimator.

(d) (1 mark) Can you argue that your choice of the estimator in the previous question is the Best Linear Unbiased Estimator (BLUE)? Show your reasoning.

(e) (1 mark) Suppose we are interested in testing $H_0 : \alpha_1 = 1.3$. Construct a linear hypothesis (i.e. find the matrix $\boldsymbol{A}$ and the vector $\boldsymbol{c}$) and show the null distribution.

(f) (1.5 marks) For ordinary least squares and the appropriate weighted least squares estimator from part (c), conduct a simulation study to estimate the probability of rejecting $H_0 : \alpha_1 = 1.3$ at significance level $\alpha = 0.05$. Identify which method has greater power.

You can use the car package in R (i.e., the linearHypothesis() function) with your specified linear hypothesis in the previous question. Note that you may access the p-value with the following option:

```
linearHypothesis(fit, hypothesis.matrix, rhs)$`Pr(>F)`[2]
```

(g) (1 mark) Is the difference in power observed in part (g) consistent with what you would expect from the results in part (c)? Explain briefly.

5. (4 points)

(a) Develop a function `myPoisson(y, x, n_iter = 20)` in R that fits a Poisson regression with the canonical link using the IRLS you derived and implemented manually (show the derivations). The function should report (i) $\hat{\beta}_{mle}$, (ii) $\widehat{\text{Cov}}(\hat{\beta}_{mle})$, and (iii) the (estimated) overdispersion $\hat{\phi}$. You may use the `glm` function to check if your algorithm is correct, but your function should be built manually.

(b) Revisit the simulation study in Lecture 4, where data are generated from (i) negative binomial and (ii) zero-inflated Poisson. Instead of fitting Poisson GLM, conduct statistical inference for $H_0 : \beta_1 = 0$ using the Wald test based on quasi-likelihood. Full credit would be given if you implement directly using the `myPoisson` function in (a), but it is okay to use `family = quasipoisson` in `glm` for some penalties. Using the same simulation setting as in the slides, report the empirical Type I error rate for both scenarios when $\alpha = 0.05$.

(c) (0.5 pt) Use negative binomial regression instead of the previous example and report the Type I error rate for both scenarios. It could be fit by using the `glm.nb` function in the `MASS` package in R, and you *don't* need to implement it yourself.